AN INJECTIVE MODULE IS INJECTIVE AS A PRESHEAF OF MODULES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument; the adjunction and the formal consequence are both inside the counted span.

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Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter sites-cohomology.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- Injective module over a ringed site and the category of presheaves of modules. Same chapter, the two categories involved and the forgetful functor between them. Supplies the objects compared and the functor with respect to which injectivity is tested.
- Adjunction between sheafification and the forgetful functor, and exactness of sheafification. Same chapter, the adjunction whose left adjoint is sheafification, and exactness of sheafification. Supplies the formal argument transferring injectivity.

Source reference. Original lemma 015Z: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 1777–1777. Navigation pointer invoked at proof line 1777; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 03CX: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 1782–1782. Navigation pointer invoked at proof line 1782; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

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Lemma 0.1. *Let $(\mathcal{C}, \mathcal{O})$ be a ringed site. An injective sheaf of modules is also injective as an object in the category $PMod(\mathcal{O})$.*

Proof. Apply Homology, Lemma 015Z to the categories $\mathcal{A} = Mod(\mathcal{O})$, $\mathcal{B} = PMod(\mathcal{O})$, the inclusion functor and sheafification. (See Modules on Sites, Section 03CX to see that all assumptions of the lemma are satisfied.) □